

Coordination of Energy Storage Deployment and Next-generation Power System Development

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Part 1

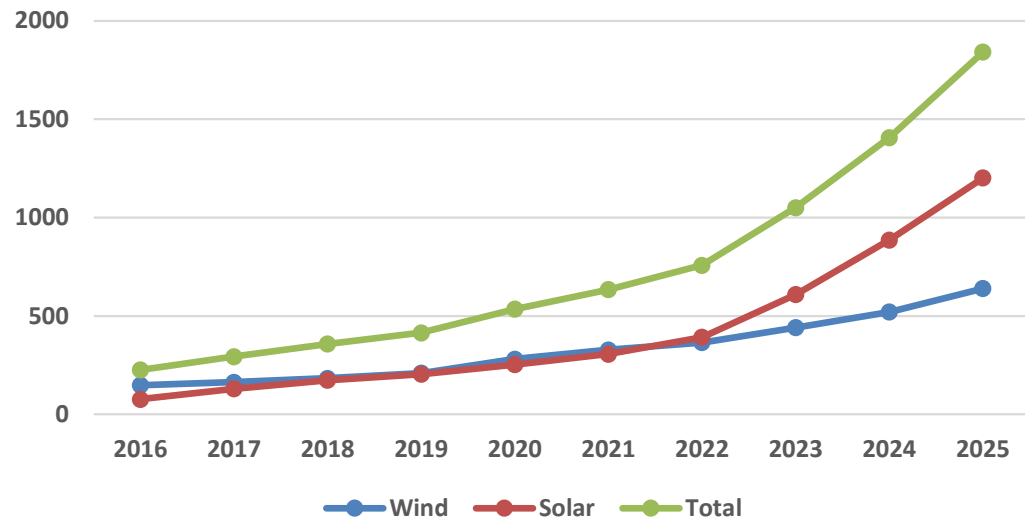
Energy Storage as a Cornerstone of the Evolving Power System



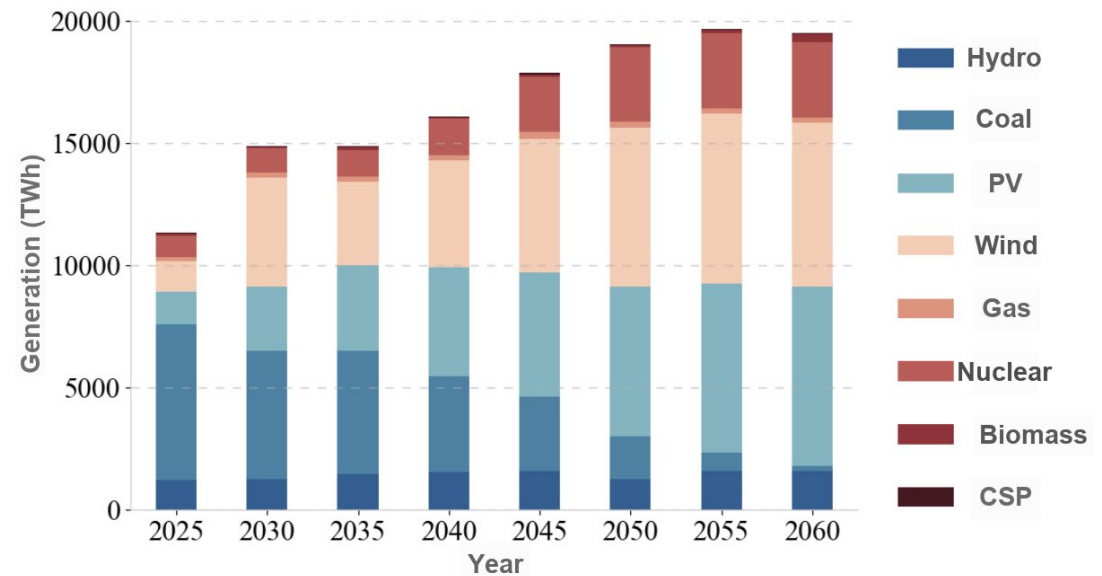
Development of Renewable Energy Generation

- Driven by both technological advances and policy support, the installed capacity of renewable generation - led by **wind and solar power** - is expanding rapidly;
- Last year, wind and solar power generation capacity added over the year exceeded **430 GW**, and cumulative installed capacity surpassed **1,800 GW** in China.
- China's cumulative installed wind and solar power capacity **surpassed** that of thermal power generation for the first time in 2025.

Accumulated Installed Capacity (GW)



Total Installed Wind and Solar Power Generation Capacity



Outlook for China's Power Generation by Source Type (2025–2060)

Load Growth and Re-electrification

- To reduce fossil fuel consumption, demand-side **electrification** is accelerating in the last decade;
- Massive new demand-side resources—electric vehicles, heating loads and distributed generations — are being connected to the power **distribution systems**;
- The distribution systems are changing from passive systems to active ones. And the distributed demand-side resources are being **aggregated** to participate power system operation.



• Electric vehicles



• Heating/cooling loads

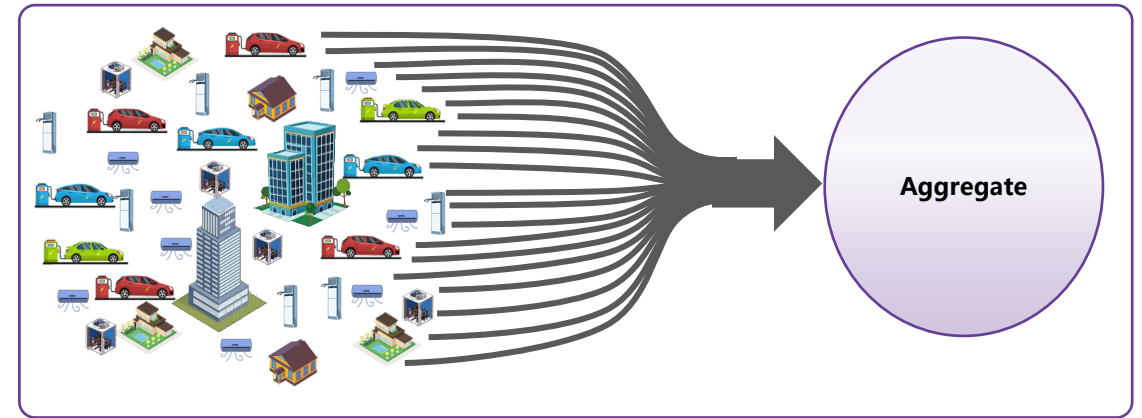


• Distributed generation



• Distributed energy storage

Integration of Massive New Demand-side Resources

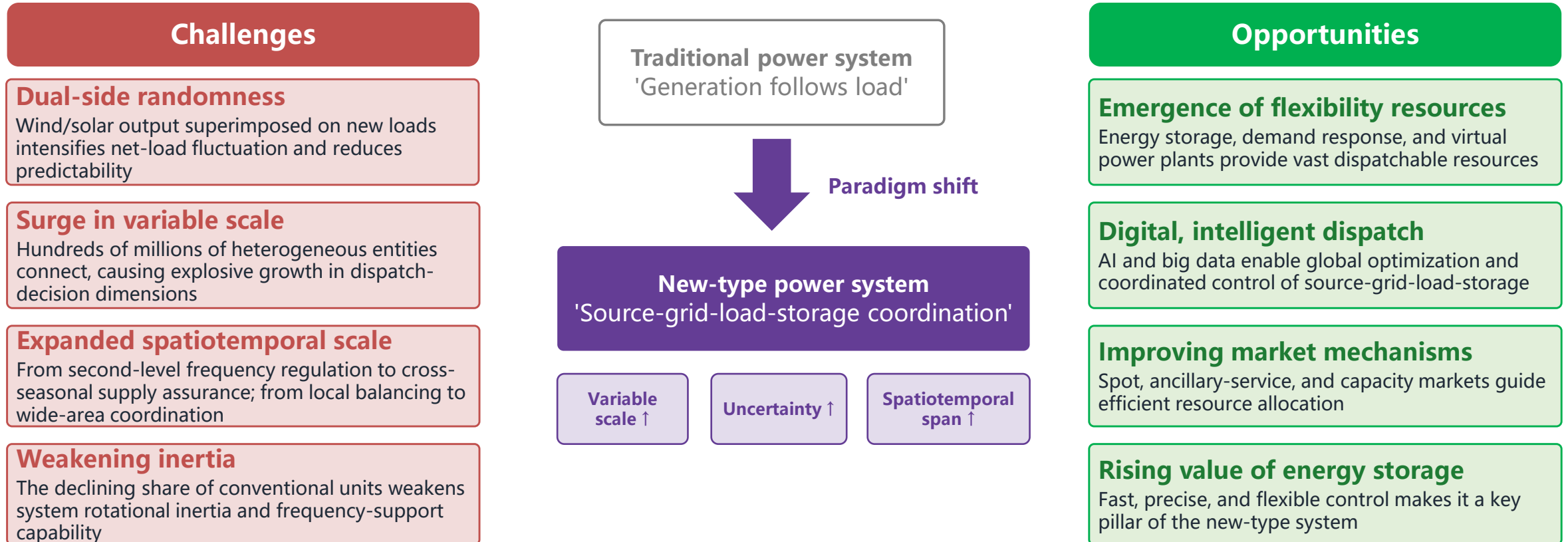


Power system
operation & dispatch

Demand-side resources participating in power-system operation

Opportunities and Challenges for Building Next-generation Power System

- The drastic changes on both the generation side and the load side of the power system have brought unprecedented challenges to its **safe and economic operation**. Renewable energy generation and load demand are highly affected by weather conditions and exhibit **strong uncertainty**, requiring new types of resources to maintain the balance between electricity generation and consumption.



The Roles for Energy Storage

- New types of energy storage, represented by lithium-ion batteries, are capable of providing support for the safe and economic operation of power systems across **multiple timescales**, and have already been widely deployed. Nevertheless, their energy coupling properties necessitate the development of **novel business models, operation and control strategies**.

Time scale	Millisecond–second	Second–minute	Minute–hour	Hour–day	Day–season
Main functions	Inertia support Primary frequency regulation Fault ride-through	Secondary frequency regulation Power smoothing	Peak shaving Ramping Spinning reserve	Peak shaving & valley filling Energy time-shift	Long-duration storage Supply assurance Seasonal balancing
Challenge addressed	Addresses low inertia and rapid frequency drops	Addresses drastic wind/solar output fluctuation	Addresses steep net-load ramps	Addresses intraday supply–demand mismatch (midday surplus, evening deficit)	Addresses renewables' seasonal imbalance
Suitable technologies	Flywheel / supercapacitor	Li-ion / flywheel	Li-ion battery	Li-ion / pumped hydro / CAES	Pumped hydro / hydrogen / CAES

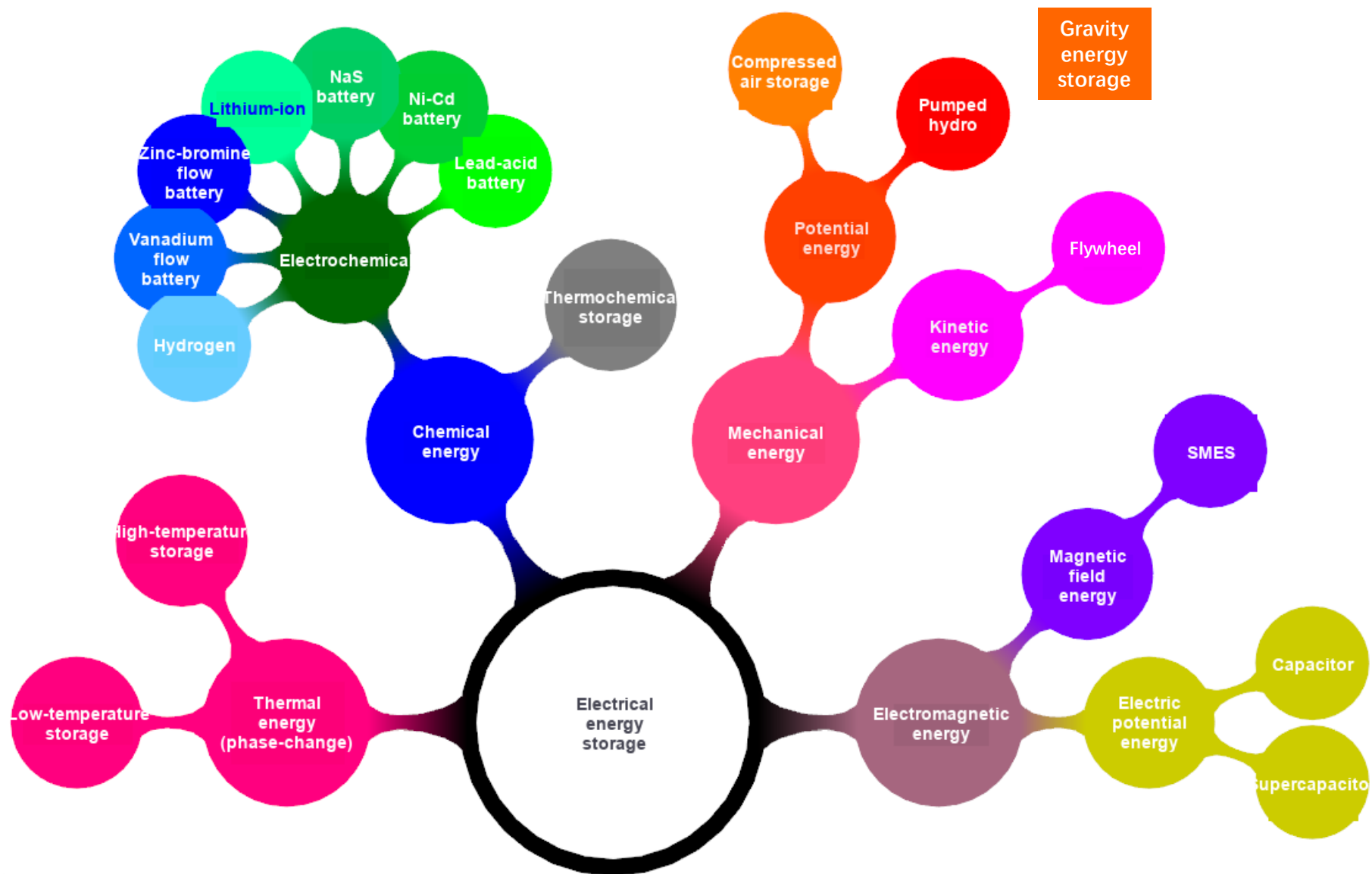


Part 2

Mainstream energy storage technologies and their characteristics

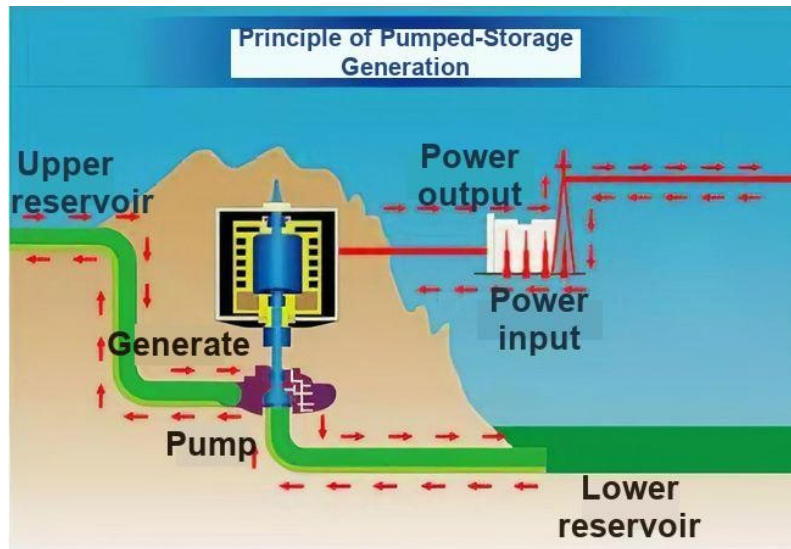


Major Categories of Energy Storage Technologies



Pumped Hydro Storage

- A pumped-hydro plant stores electricity as **the potential energy** of water and, when needed, releases water from the upper reservoir to generate power;
- The technology is **mature**, with a service life of over 50 years, a high cycle count, and installed capacities reaching millions of kW;
- Pumped-hydro plants have demanding siting requirements and long construction periods, with a power cost of about **4,000–7,000 RMB/kW**.



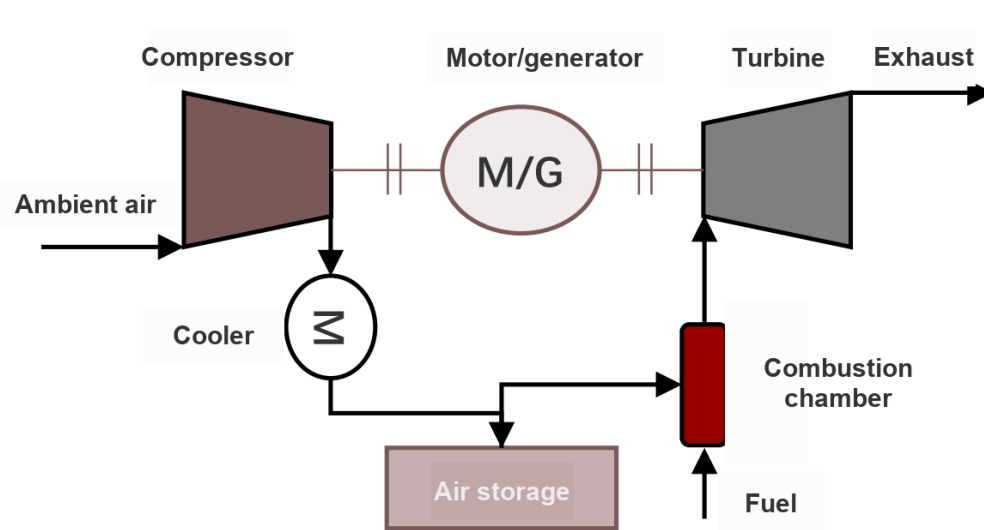
1. Use surplus grid power to drive the pump.
2. Pump water from the lower to the upper reservoir to store it.
3. Release water to generate during grid peak/near-peak; water flows to the reservoir.
4. Pump four, generate three (4 kWh to pump, 3 kWh generated).

2021 Approved Pumped-Storage Station Cost Levels

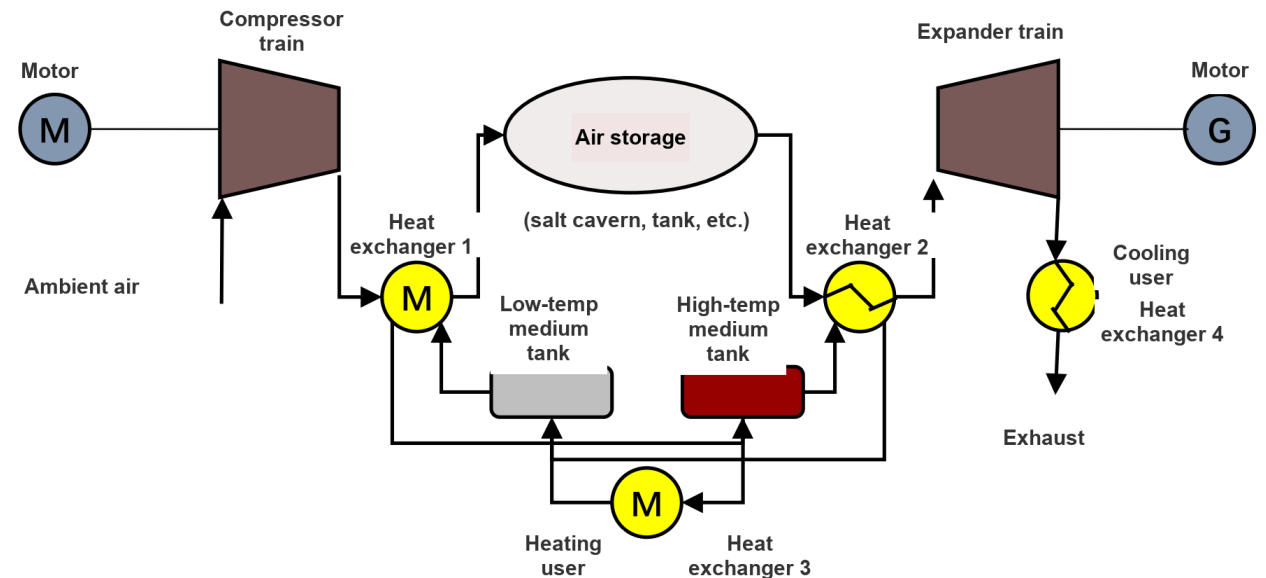
Province	Project	Capacity (10k kW)	Unit static (RMB/kW)	Unit dynamic (RMB/kW)
Heilongjiang	Shangzhi	120	5,809	6,965
Zhejiang	Taishun	120	4,888	5,945
	Tiantai	170	5,263	6,319
Jiangxi	Fengxin	120	5,298	6,366
Henan	Lushan	130	5,590	6,675
Hubei	Pingtanyuan	140	5,495	6,720
Chongqing	Liziwan	140	5,828	7,260
Guangxi	Nanning	120	5,476	6,613
Ningxia	Niushoushan	100	6,364	7,847

Compressed Air Energy Storage

- Compressed air energy storage (CAES) offers large storage capacity, long storage duration, long operating life, and **low kWh investment**.
- Conventional CAES systems have no thermal-storage subsystem, so the heat of compression is discarded, resulting in **low efficiency** (around 50%) and consumption of natural gas;
- Advanced CAES incorporates **thermal storage**, eliminating the need to burn fossil fuels during discharge; the combustion chamber can be removed, improving efficiency.



Combustion CAES

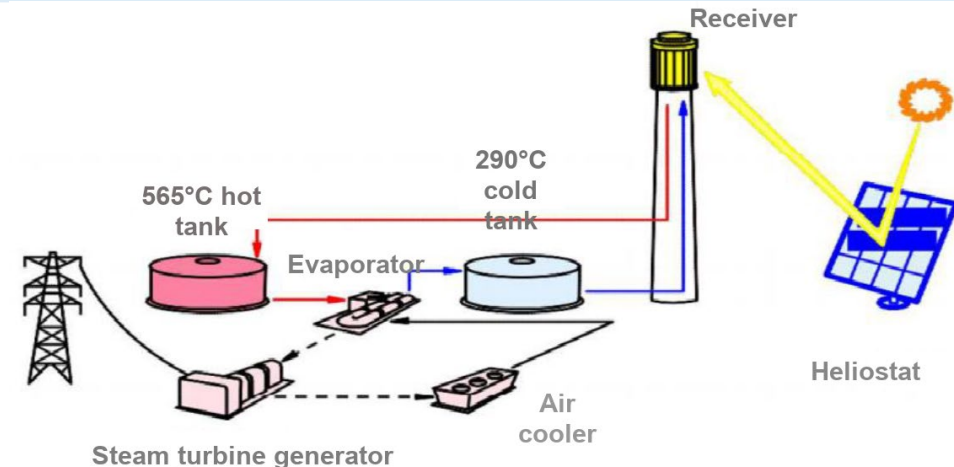


Non-combustion CAES



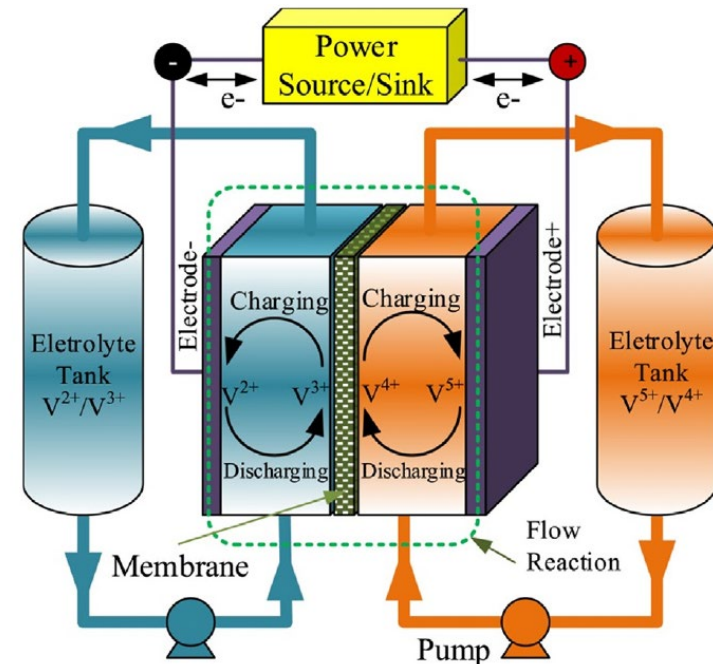
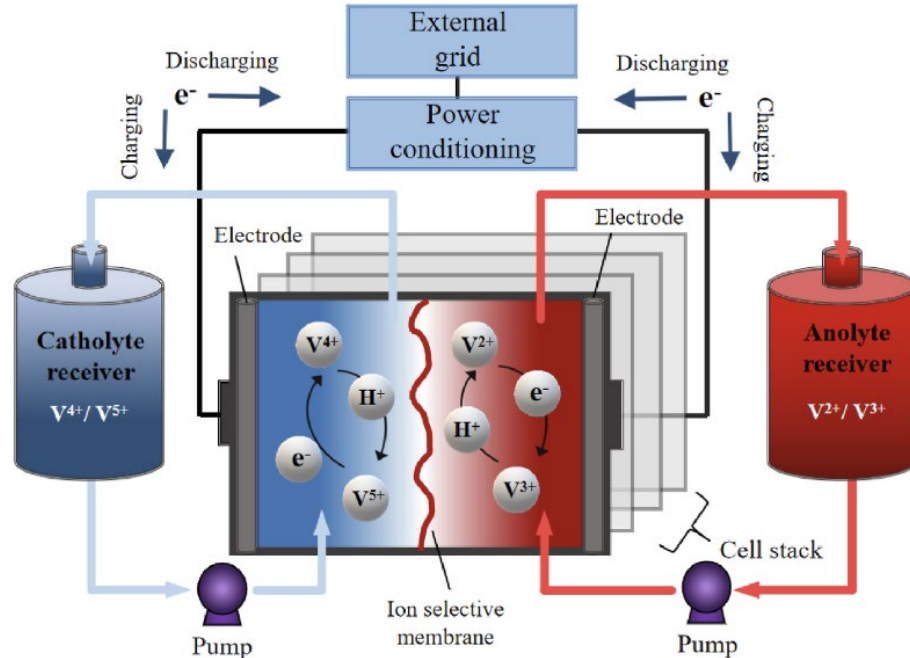
Concentrated Solar Power Plant and Molten-Salt Storage

- Principle of concentrated solar power (CSP): **concentrators** such as mirrors and lenses focus collect solar radiation onto a receiver, heating a heat-transfer medium such as molten salt or thermal oil; through a heat exchanger, this medium heats water into high-temperature, high-pressure steam that drives a steam turbine coupled to a generator.
- **Molten salt** operates from about 240 °C to about 560 °C with low vapor pressure, making it suitable for large-scale thermal storage. The molten salt output by the storage system can steadily produce high-temperature, high-pressure steam of around 550 °C via a heat-exchange system to drive the turbine and generate power.
- Molten-salt storage systems used in CSP can reach capacities of the **GWh scale** and above, with excellent thermal insulation—temperature drops by less than 2 °C over 24 hours.



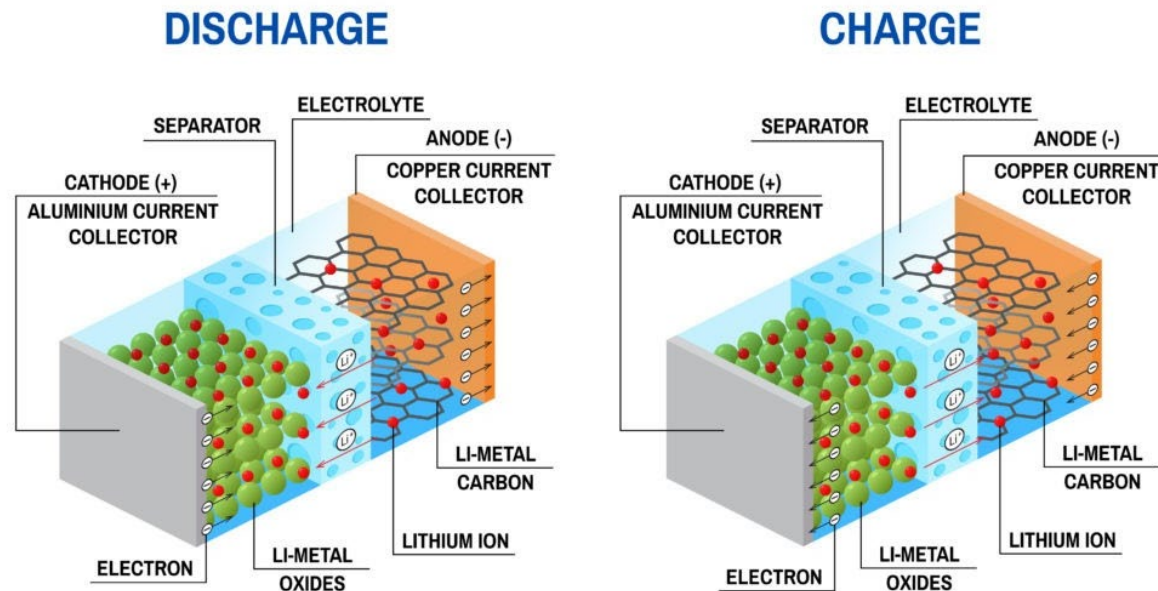
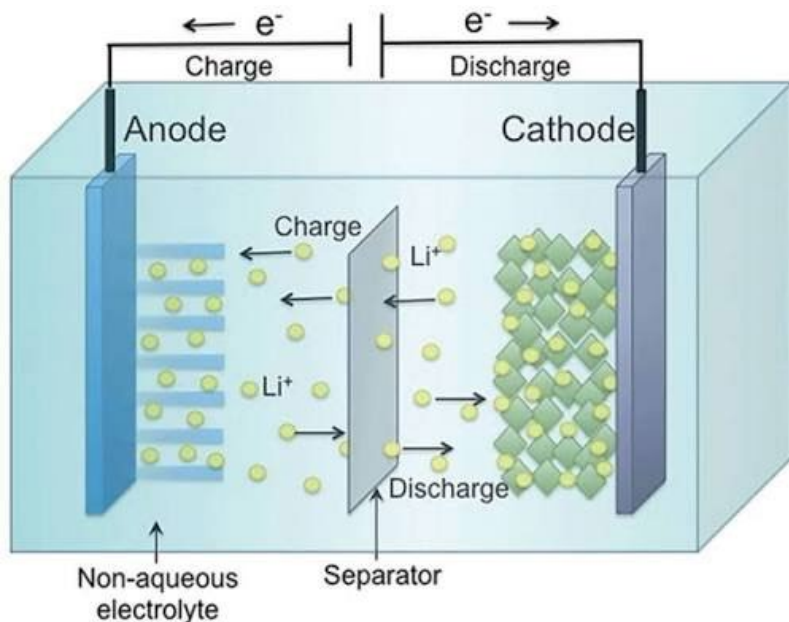
Flow Battery Energy Storage

- Flow batteries offer high safety, **long life, and large scale**, giving them strong prospects in large-scale storage. The U.S. DOE projects that flow-battery storage will have a comparative advantage in grid-scale storage with durations of 4–10 h.
- The first phase of the Dalian **100 MW/400 MWh** all-vanadium flow-battery storage station started operation in 2022; by the end of 2025, the **200 MW/1000 MWh** vanadium flow-battery storage station of Three Gorges Energy's Jimsar solar-plus-storage project in Xinjiang achieved full-capacity grid-connected operation.



Lithium Battery Energy Storage

- Lithium-ion batteries offer **high energy density, high charge/discharge efficiency, fast response**, and a complete industrial chain, making them the fastest-growing electrochemical storage technology in recent years.
- By the end of 2025, the cumulative installed capacity of commissioned battery-storage projects worldwide reached about **315 GWh**, up nearly 50% year on year.
- China and the United States lead deployment, with China far ahead of all other markets.



Hydrogen Energy Storage

- Hydrogen storage is anticipated to become an important option for **long-duration, large-scale storage** and is key to carbon neutrality in non-electric energy consumption.
- Hydrogen storage can store surplus electricity from intermittent renewables such as solar and wind—or power that cannot be connected to the grid—by producing hydrogen via water electrolysis. This hydrogen can be consumed locally, used to generate power through **fuel cells**, or transported by pipeline or tube trailer to downstream end users.

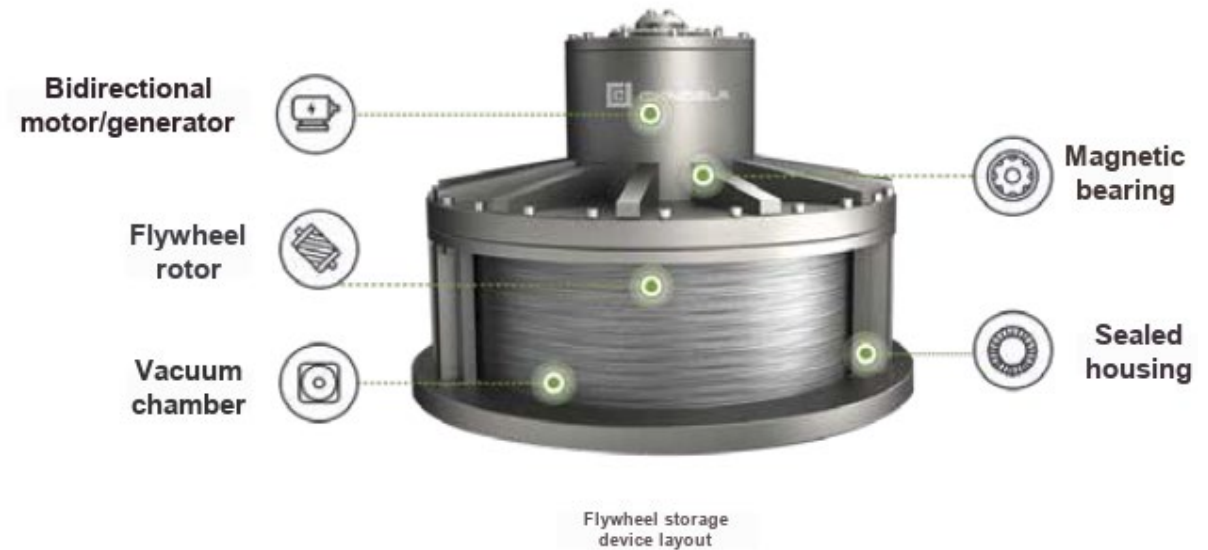
Parameter	Battery			Hydrogen
	LFP battery	Na-ion battery	Flow battery	
Energy density (MJ/kg)	0.5	0.4	0.4	140
Round-trip efficiency	85%	85%	80%	36%
Monthly self-discharge	Varies with temperature and humidity; generally below 10%			None
Cycle life (cycles)	3000	2000	>10000	None
Initial investment cost (RMB/kWh)	1500	1200	2500	1300

Comparison of mainstream electrochemical storage technologies

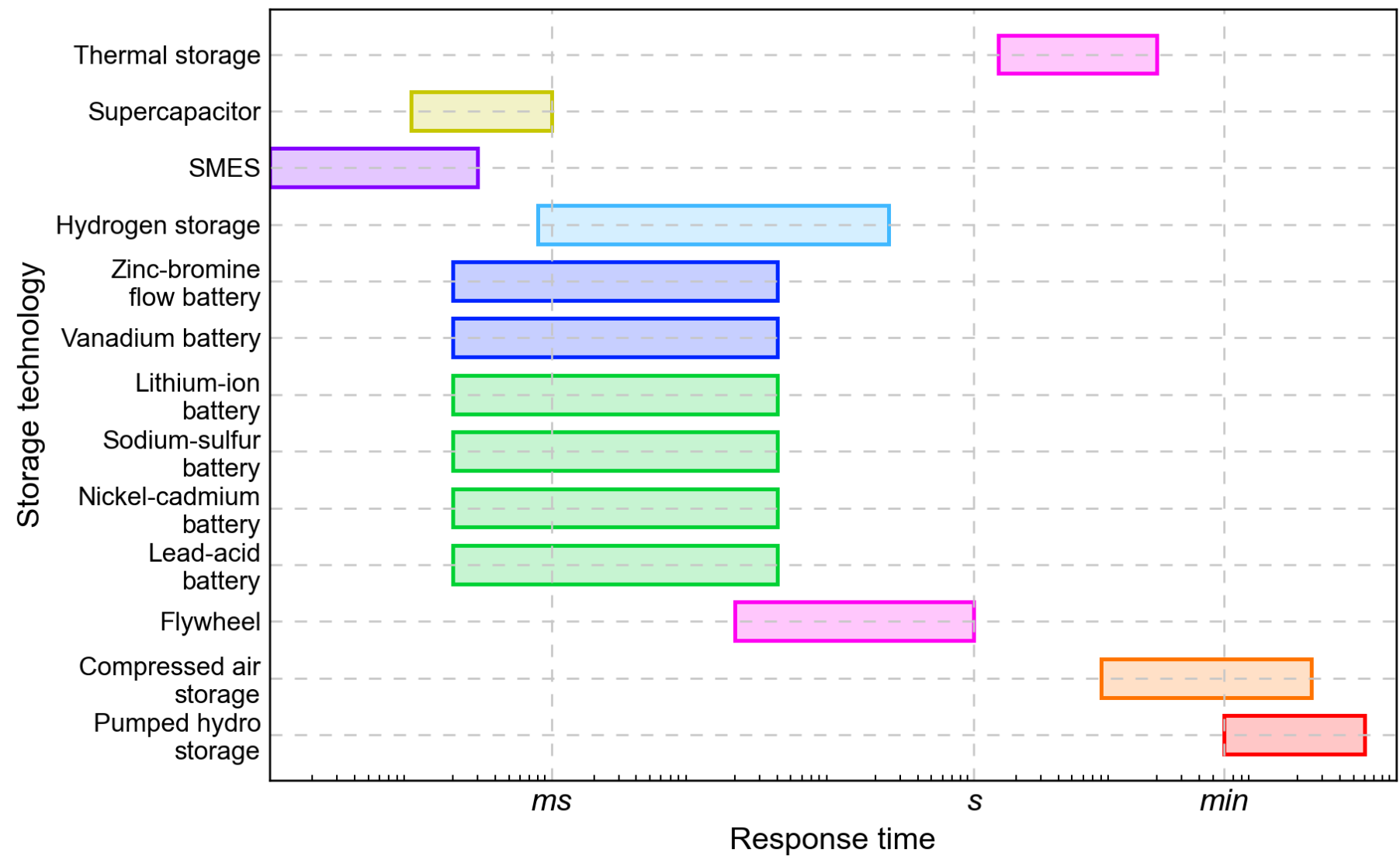


Flywheel Energy Storage

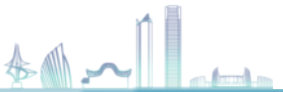
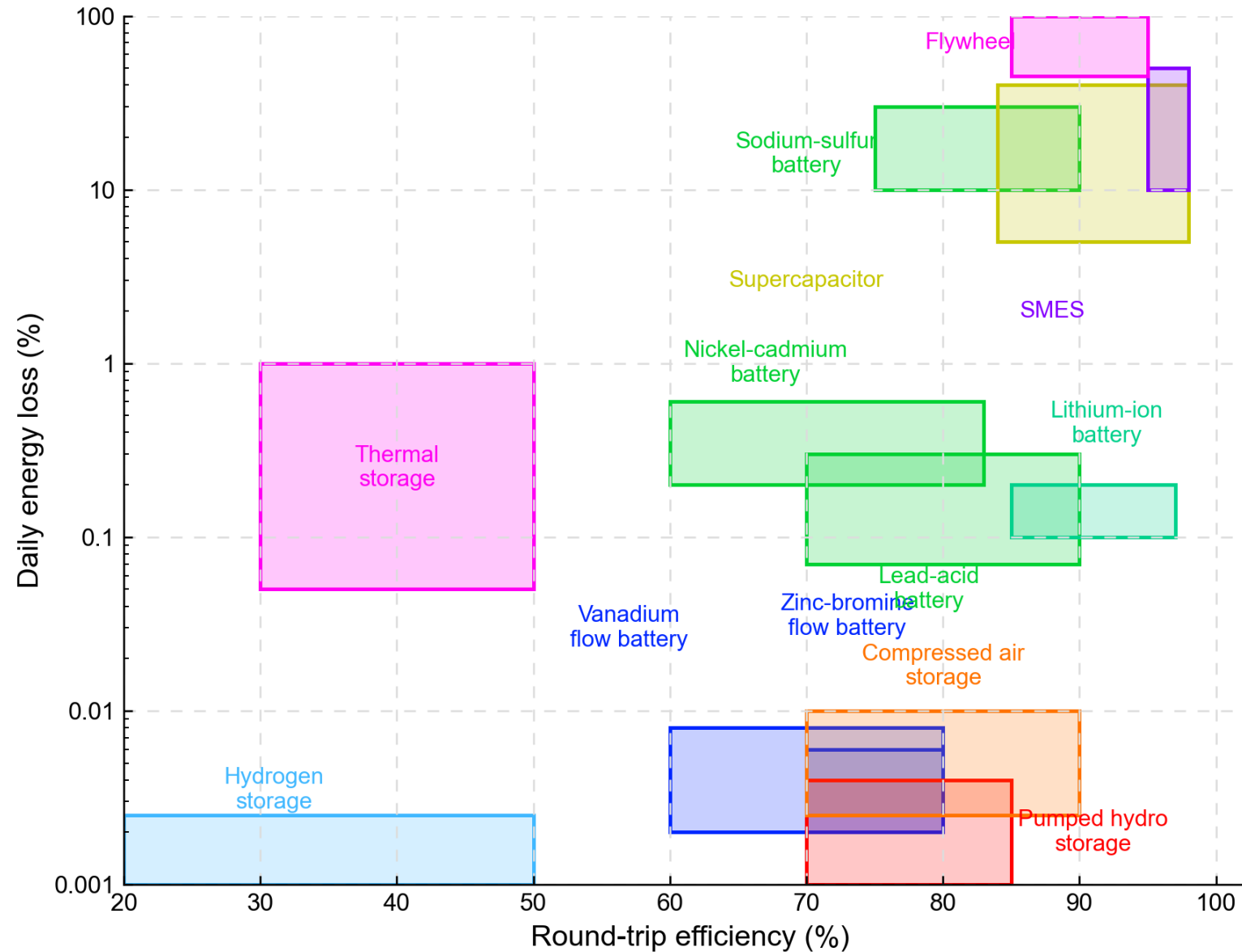
- Flywheel storage features relatively **high power density, a high number of charge/discharge cycles**, low operating-environment requirements, and no pollution.
- Its conversion efficiency is high, exceeding 90%, with a service life of up to 25 years; however, continuous discharge **lasts only minutes** and energy density is low. It is a typical **power-type storage technology** with strong prospects in short-duration, high-frequency applications.



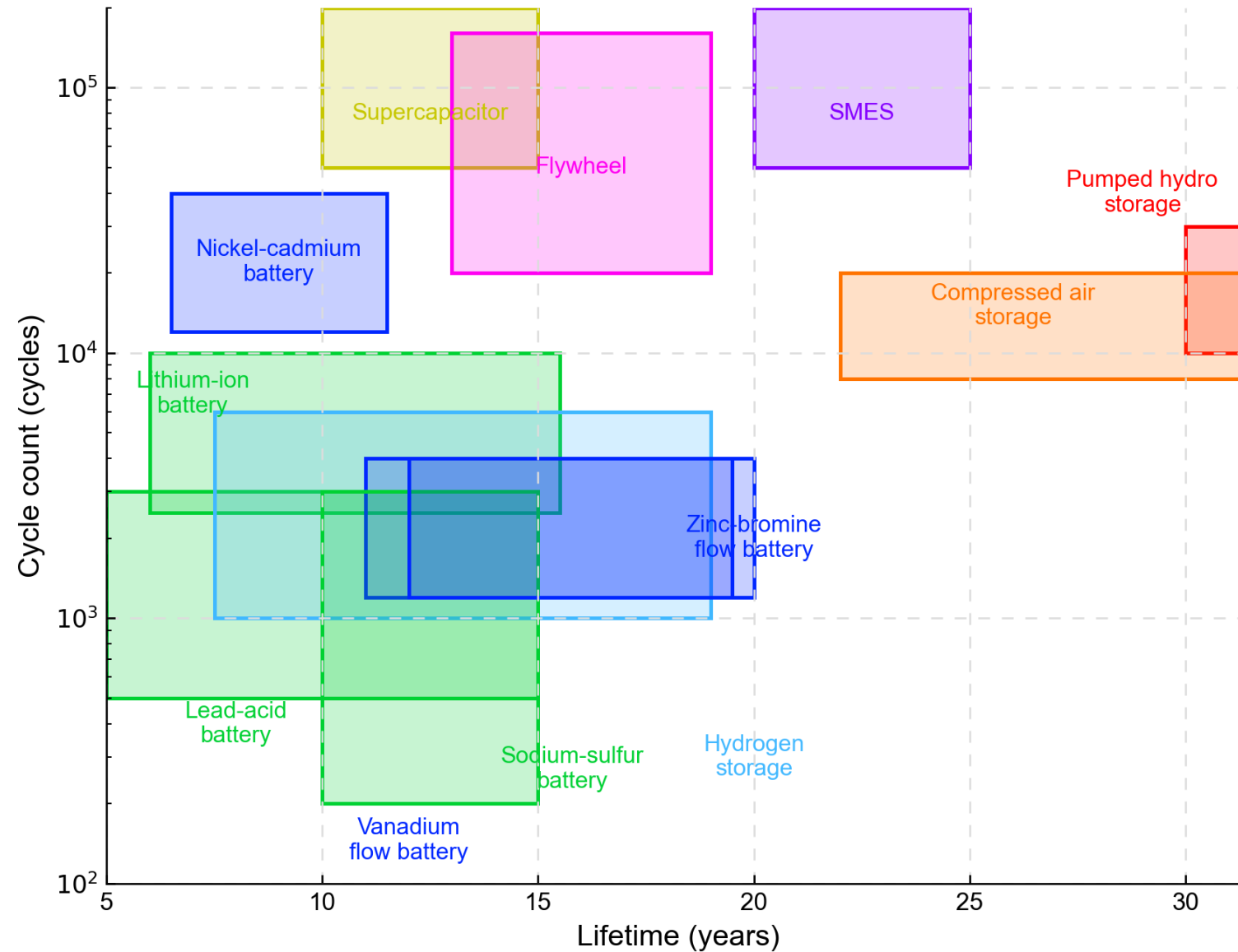
Comparison of Energy Storage Technologies — Response Time



Comparison of Energy Storage Technology — Efficiency

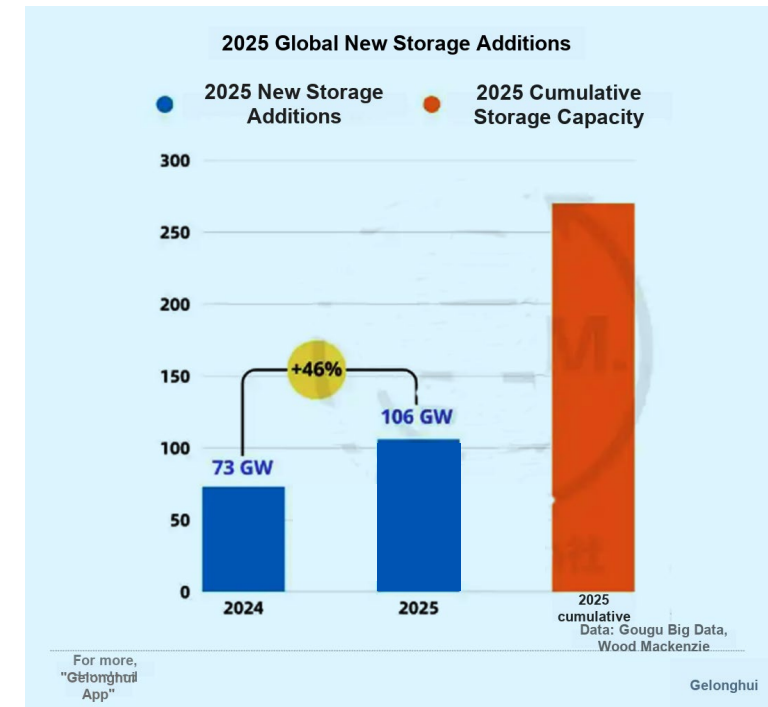


Comparison of Energy Storage Technology — Lifetime



Global Electricity Storage Project Scale (as of end-2025)

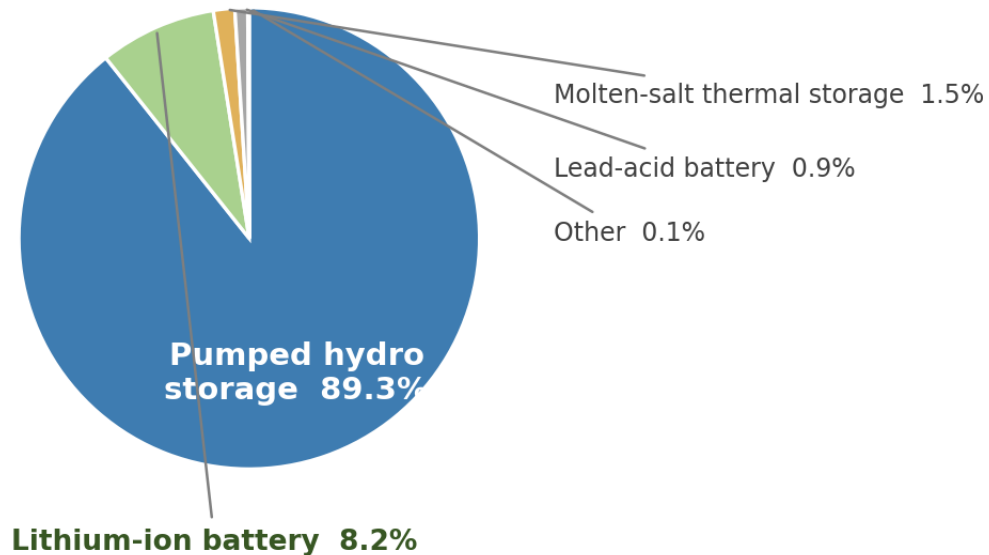
- By the end of 2025, the cumulative installed capacity of commissioned electricity-storage projects worldwide reached **496.2 GW**, up 33.3% year on year. Pumped hydro's share of cumulative capacity fell below **50%** for the first time, down 11.5 percentage points from the same period in 2024.
- This indicates that the global storage technology mix is accelerating toward diversification. New-type storage maintained rapid growth, with cumulative installed capacity reaching **278.7 GW/687.5 GWh**.



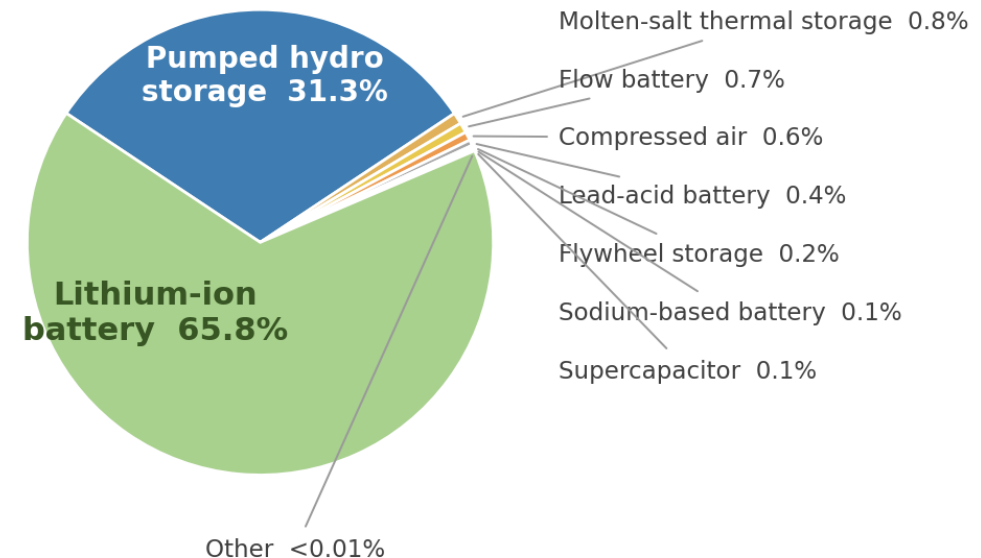
China's Electricity Storage Project Scale (as of end-2025)

- China's cumulative commissioned electricity storage reached **213.3 GW**, up 54% year on year.
- New-type storage, led by lithium batteries, achieved leapfrog growth; its share of installed capacity exceeded **two-thirds**, accelerating the shift from a single technology to diversification.

As of the end of the “13th Five-Year Plan”



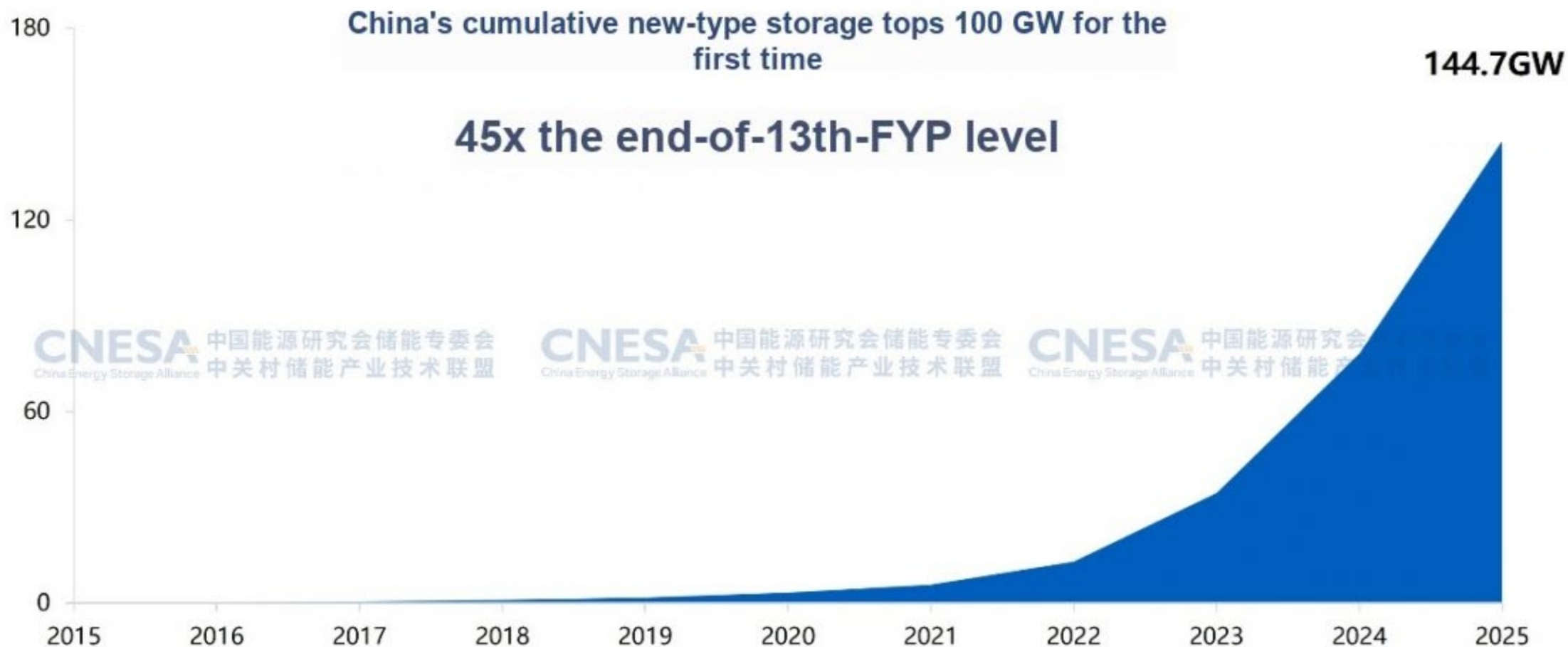
As of the end of the “14th Five-Year Plan”



Data source: CNESA



China's Electricity Storage Project Scale (as of end-2025)



Data source: CNESA



Part 3

Application of Energy Storage Across All Sectors of New-generation Power Systems

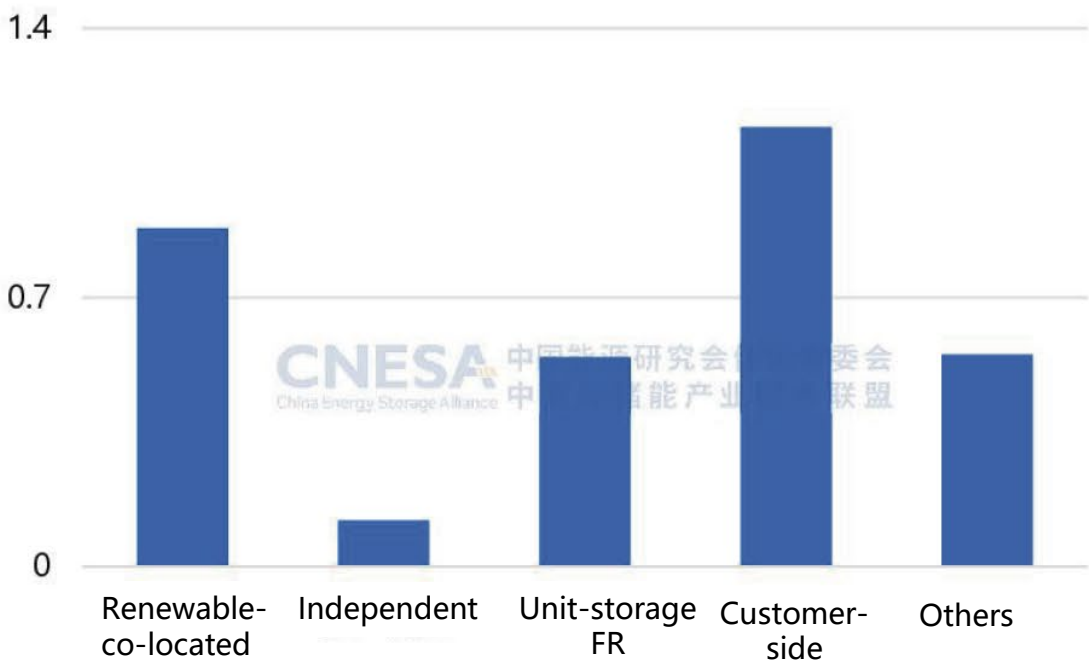


Application Scenarios of China's Energy Storage Projects

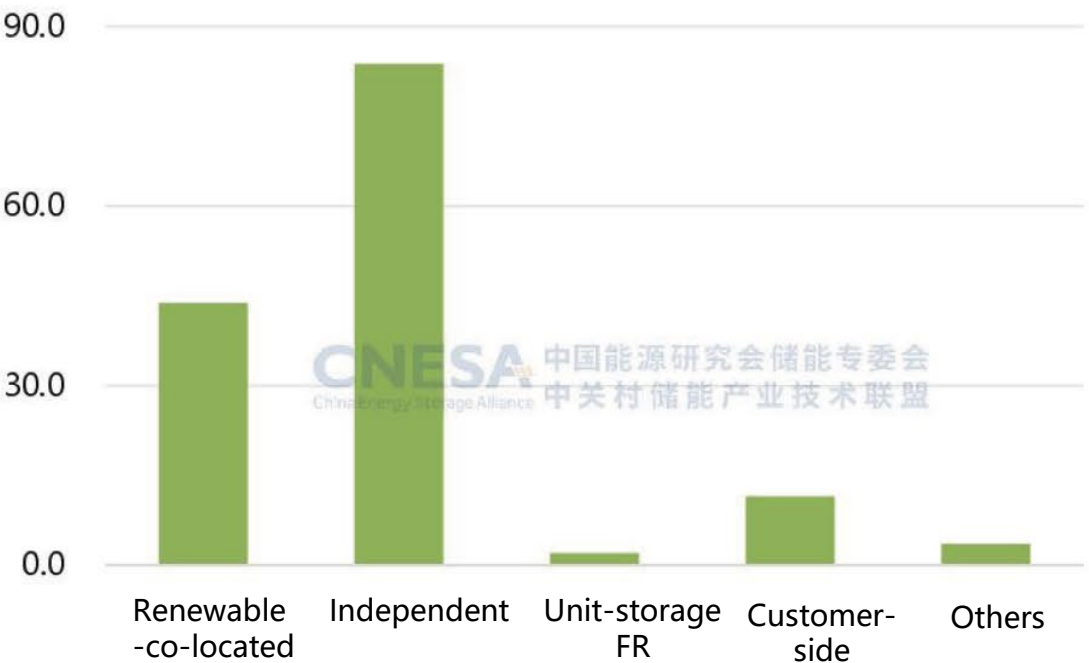
- Application mix has shifted: customer-side storage, formerly dominant at 35%, has been replaced by **independent energy storage at 58%**. Thermal-storage frequency regulation (1.4%) and customer-side storage (8%) declined markedly, while renewable co-located storage remained stable.

China New-Type Storage by Application, Cumulative (GW)

End of 13th FYP



End of 14th FYP



Data source: CNESA



3.1 Energy Storage Deployed at Renewable Power Plants

Output smoothing

- Meet grid-connection standards
- Research focus: filtering algorithms

Reduce wind/solar curtailment

- Absorb curtailed energy when constrained; release when unconstrained
- Research focus: charge/discharge strategy

Voltage regulation

- Enhance reactive-power support
- Research focus: coordinate with AVC reactive-power compensation

Participate in grid operation

- Participate in the energy market; provide ancillary services
- Research focus: charge/discharge scheduling & control strategy



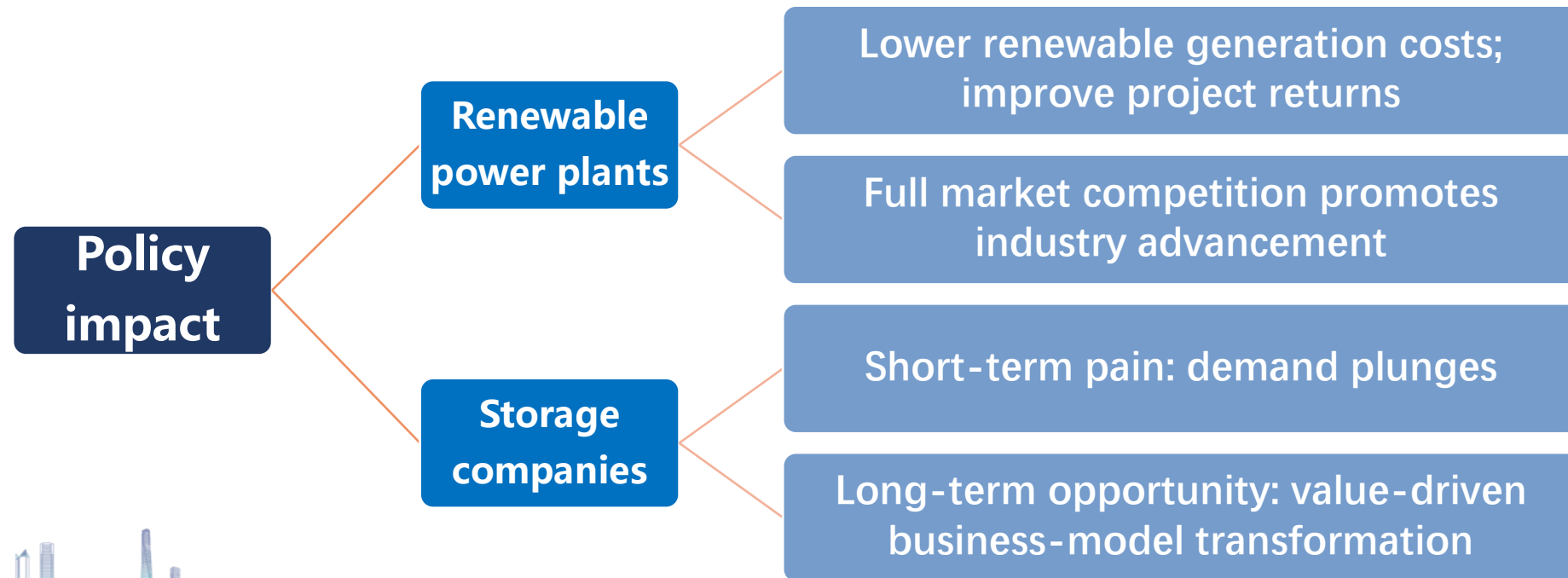
Status of Energy Storage Installation at Renewable Power Plants

➤ By the end of 2024, China's renewable-co-located storage capacity was about **30.97 GW/73.79 GWh** (i.e., about 31 GW of power and about 74 GWh of energy), accounting for roughly **42%** of total new-type storage installed capacity.

Feature	Description
Installed capacity	Renewable-co-located storage (wind/solar-plus-storage) is a major component of new-type storage; by the end of 2024 it accounted for nearly half of installed capacity and, together with independent/shared storage, forms a storage system dominated by lithium-ion batteries.
Regional distribution	Mainly located in Northwest and North China , where large renewable bases are concentrated; the co-location ratio exceeds 50% in both regions.
Single-station scale & duration	Most co-located storage stations range from 10 MW to 100 MW (about 46%) or above 100 MW (about 49%). Their storage duration is mostly 2–4 hours.
Development status	Northwest China, North China, Shandong and other regions require renewable plants to allocate storage at roughly 10%–20% . In 2024, the utilization hours and charge/discharge cycles of renewable-co-located storage improved markedly.

Status of Energy Storage Installation at Renewable Power Plants

- On February 9, 2025, Chinese government issued the 'Notice on Deepening Market-Oriented Reform of Renewable Feed-in Tariffs to Promote High-Quality Development of Renewables' (**Document No. 136**), explicitly stating that 'installing energy storage must not be a precondition for the approval, grid connection, or power generation of new renewable projects.' This marks the official end of the mandatory renewable-storage-allocation policy that had been in place for nearly **eight years**.



3.2 Development of Independent Energy Storage Stations

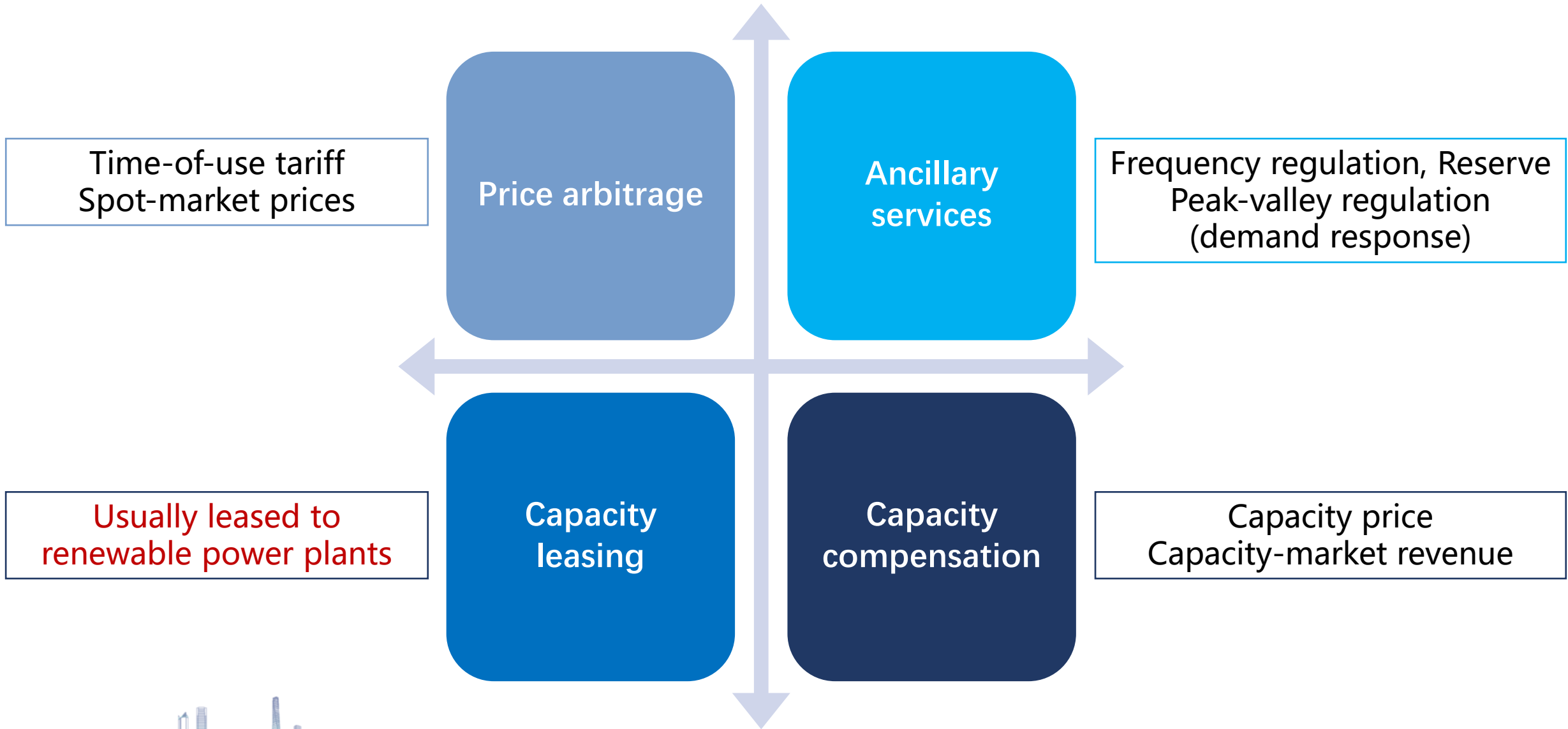
- In 2025–2026, China's independent storage stations (including shared storage) have become the main driver of new-type storage growth, with their cumulative share **exceeding 50%**. Their role has shifted from 'appendage' to independent market entity; through a diversified revenue model of '**peak-valley arbitrage + ancillary services + capacity compensation**', they are transitioning from policy-driven to market-based operation, with clear trends toward larger single stations (above 100 MW) and longer durations (2–4 h).

By the end of 2025, independent storage accounted for more than half of all new-type storage, with 35.43 GW of newly added capacity; the development focus has shifted from pure renewable co-location toward grid-side regulation resources centered on independent storage.

Scale and technical characteristics	
Larger single stations	Projects of 100 MW and above account for over 70% , a significant increase
Storage duration	2–4 hour projects dominate the market, while projects over 4 hours are growing rapidly
Technology pathway	Lithium-ion batteries still dominate (over 96%), while long-duration technologies such as compressed air are beginning to be deployed



Revenue of Independent Energy Storage Stations



3.3 Deployment of Distributed Storage in Distribution Networks

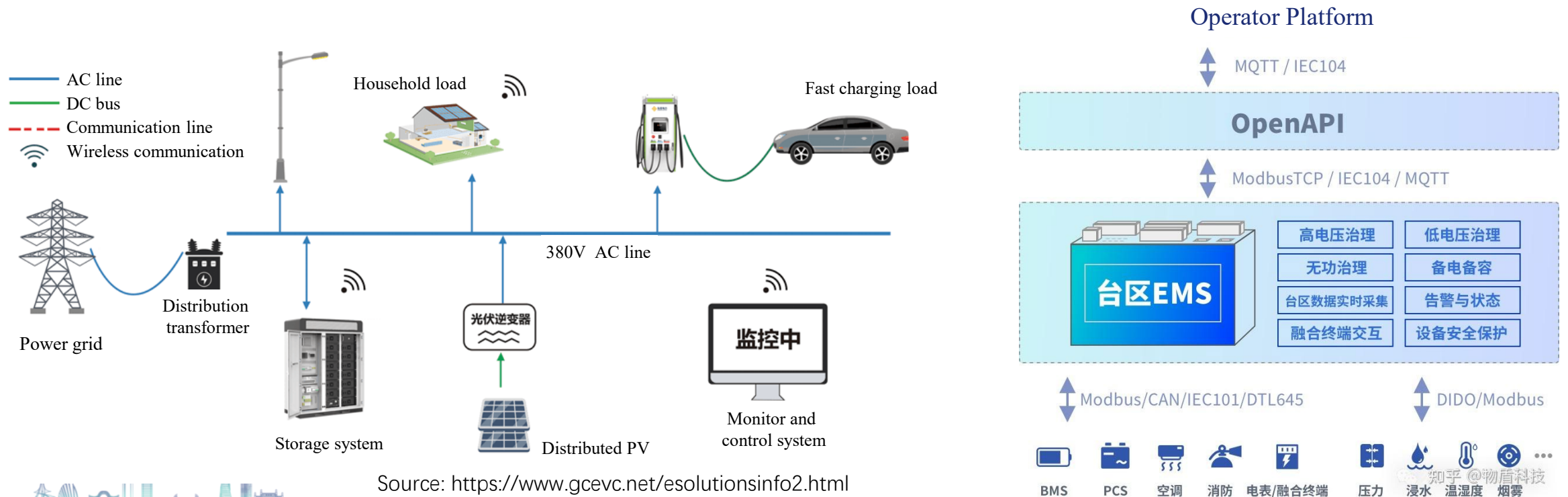
- Distributed energy storage refers to electrical storage systems connected to the public distribution network, which can be either **stationary or mobile storage systems**.

Roles	Description	Illustration
Energy Storage for Supporting Distribution System	➤ Deployed in distribution line or besides distribution transformer ➤ Support economic and secure operation of distribution network ➤ Increasing capacity for DG integration	
Energy Storage Coordinated with DG	➤ Promote the DG integration capacity ➤ Increase the revenue of DG ➤ Improve power quality	
Mobile Energy Storage	➤ Promote the DG integration capacity ➤ Support economic and secure operation of distribution network ➤ Emergency power supply and recovery	



Deployment of Energy Storage Beside Distribution Transformer

- The deployment of **BESS beside distribution transformer** is a hot topic and increasing very fast in recent years mainly because of the increasing penetration of PV and charging facilities.
- The owner can be distribution system (DS) operator and third-party investor. The challenges include the coordinated operation between the DS and the BESS.
- The support of BESS to DS should be properly compensated, which is still an open problem.



Application of Mobile Energy Storage Systems

- Mobile energy storage system (MESS)

➤ **Development trend:** Bigger capacity, higher performance, greater adaptability



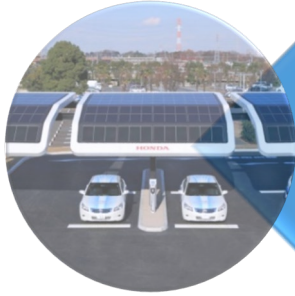
Role	Advantages	Applications
Decouple supply & demand in time & space	Mobile · Flexible · Plug-and-play · Wide-area distribution · Adaptive	• Emergency supply • Event backup • Mobile charging • Load smoothing • DG hosting

- Research Significance

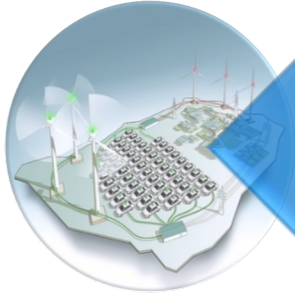
- Boost DER hosting & grid balance via optimized multi-MESS dispatch
- Enable grid transformation & secure operation for power systems
- Expand MESS applications across multi-scenario and multi-stage contexts, advancing dispatch optimization research considering high-penetration DERs in multi-distribution networks.



3.4 Demand-Side Applications of Energy Storage



Storage at EV charging stations



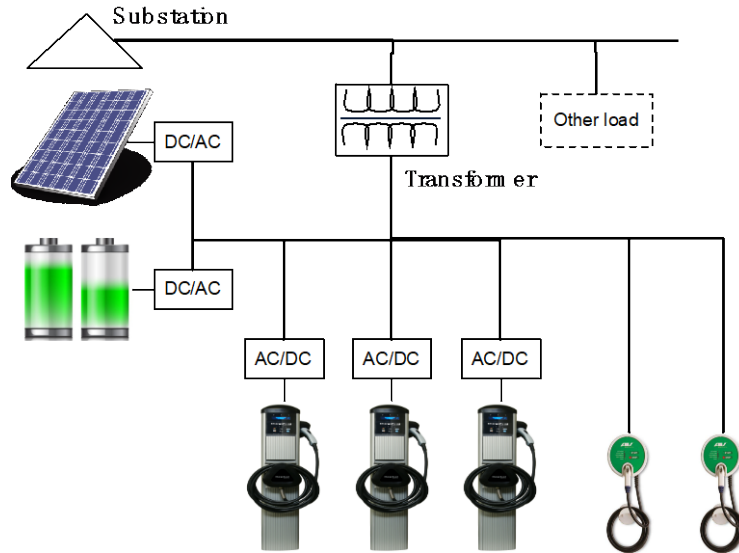
Storage in industrial parks & commercial buildings



Storage in communities or homes



Energy Storage at Charging Stations



- Reduce **the required connection capacity** of charging stations
- Reduce demand fees and enable price arbitrage
- Maximize the consumption of the electricity generated by local DG (PV)
- Smooth charging loads
- Improve power quality

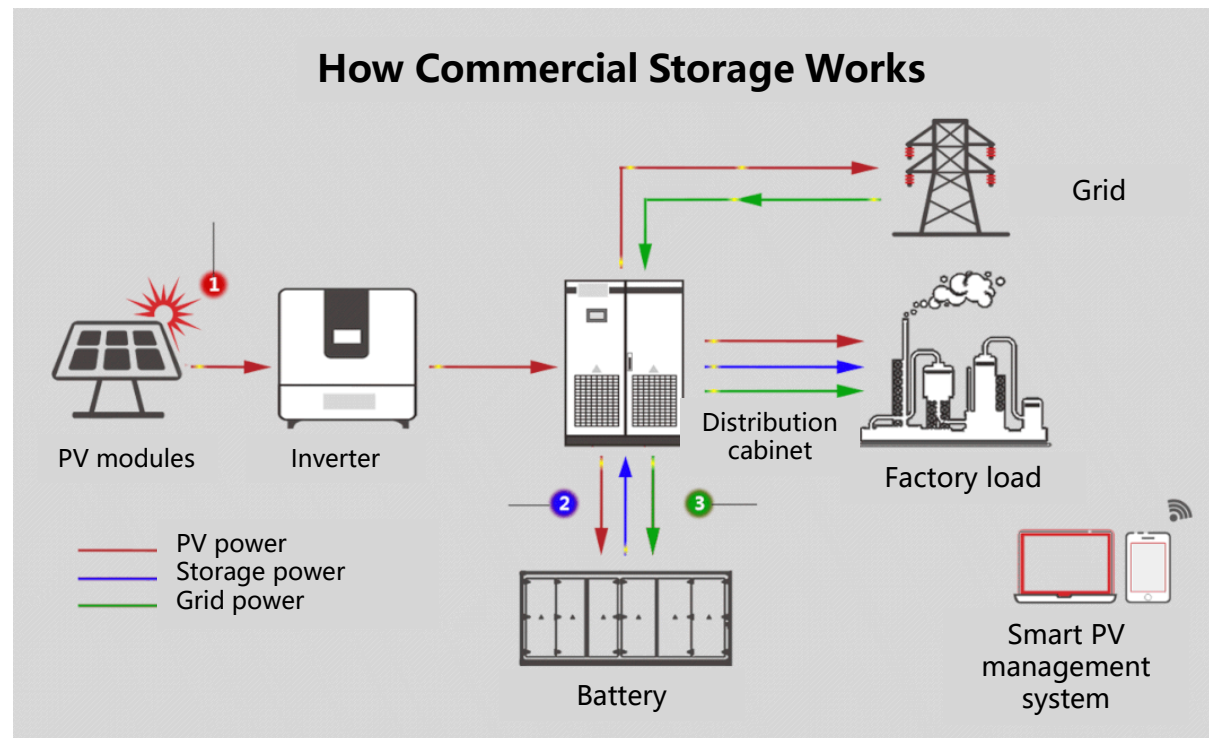


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Energy Storage in Industrial Parks and Commercial Buildings

- Energy storage in industrial parks and commercial buildings takes many forms; considering cost-effectiveness and other factors, options include **ice thermal storage, phase-change storage, or battery storage**.
- In industrial parks where PV can be installed, PV and battery storage can be deployed together to achieve goals such as **price arbitrage and improved power quality**.



Source: <https://zhuanlan.zhihu.com/p/614594787>

Residential Energy Storage

❑ Multiple products already available

- Many products are exported from China

❑ Residential storage depends mainly on electricity prices and PV

- Applied in regions with high electricity prices
- Mostly paired with PV

❑ Operating strategies for residential storage

- Set charging/discharging by electricity price
- Self-learning capability
- Joint optimization when PV is present

• Tesla's residential Powerwall

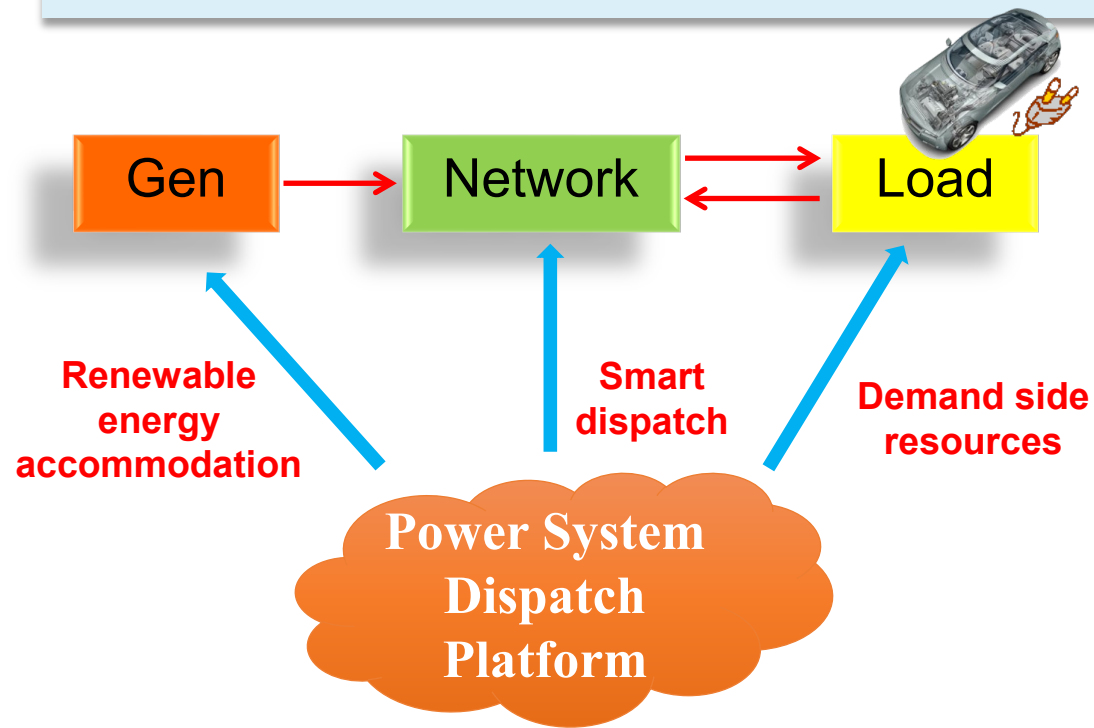
https://en.wikipedia.org/wiki/Tesla_Powerwall

• PowerVault, launched by a UK company (<http://www.powervault.co.uk/>)



3.5 Large-Scale EV Development and Storage Resources

- EVs are a new type of **load / distributed storage resource** with flexibility/controllability. They can be incorporated into demand-side management, grid dispatch, and **coordination with renewable generation**, thereby could be an important player in power-system operational optimization and renewable-energy accommodation.

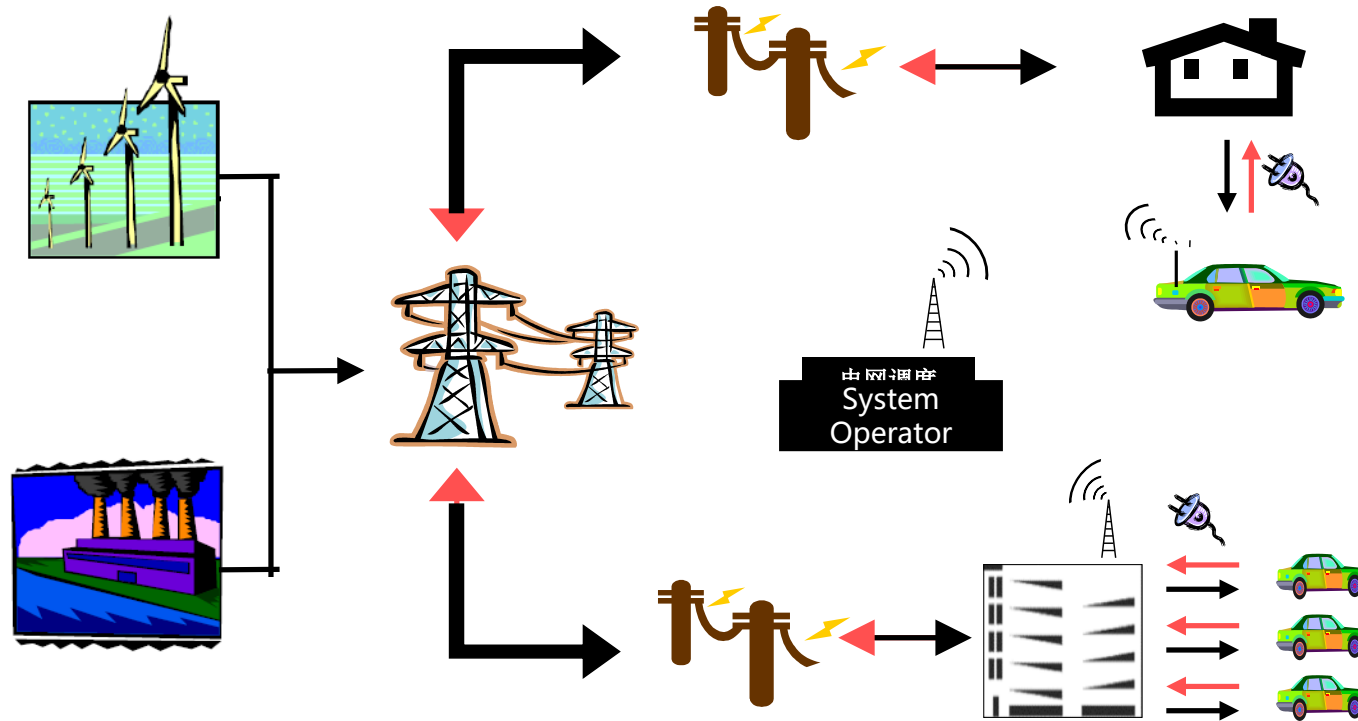


EVs can be mobile distributed storage resources

- **EV smart (orderly) charging**
 - Participates in grid regulation as a **controllable load**, with charging time and charging power as the control means.
- **Bidirectional vehicle-grid interaction (Vehicle-to-Grid, V2G)**
 - Vehicle-grid interaction means EVs act as **distributed storage units**, participating in grid regulation through charging and discharging.

Vehicle-grid Interaction (V2G)

- Vehicle-grid interaction or vehicle to grid (V2G) aims to leverage the storage capability of EVs through **bidirectional** electricity flow to support the economic and secure operation of power system;
- The basic premise of V2G is that it does not affect the use of EV, or that the user receives an acceptable return/compensation.



Arrow direction indicates the direction of power flow

Source: Kempton et al., 2001

Vehicle-grid Interaction (V2G)

➤ Major functions of V2G

Peak shaving & valley filling

- Charge during load valleys, discharge during peak load
- Extension: coordinate with renewable output

Frequency regulation

- Participate in primary frequency regulation (local control)
- Participate in secondary frequency regulation (incorporated into AGC the AGC system)

Voltage regulation

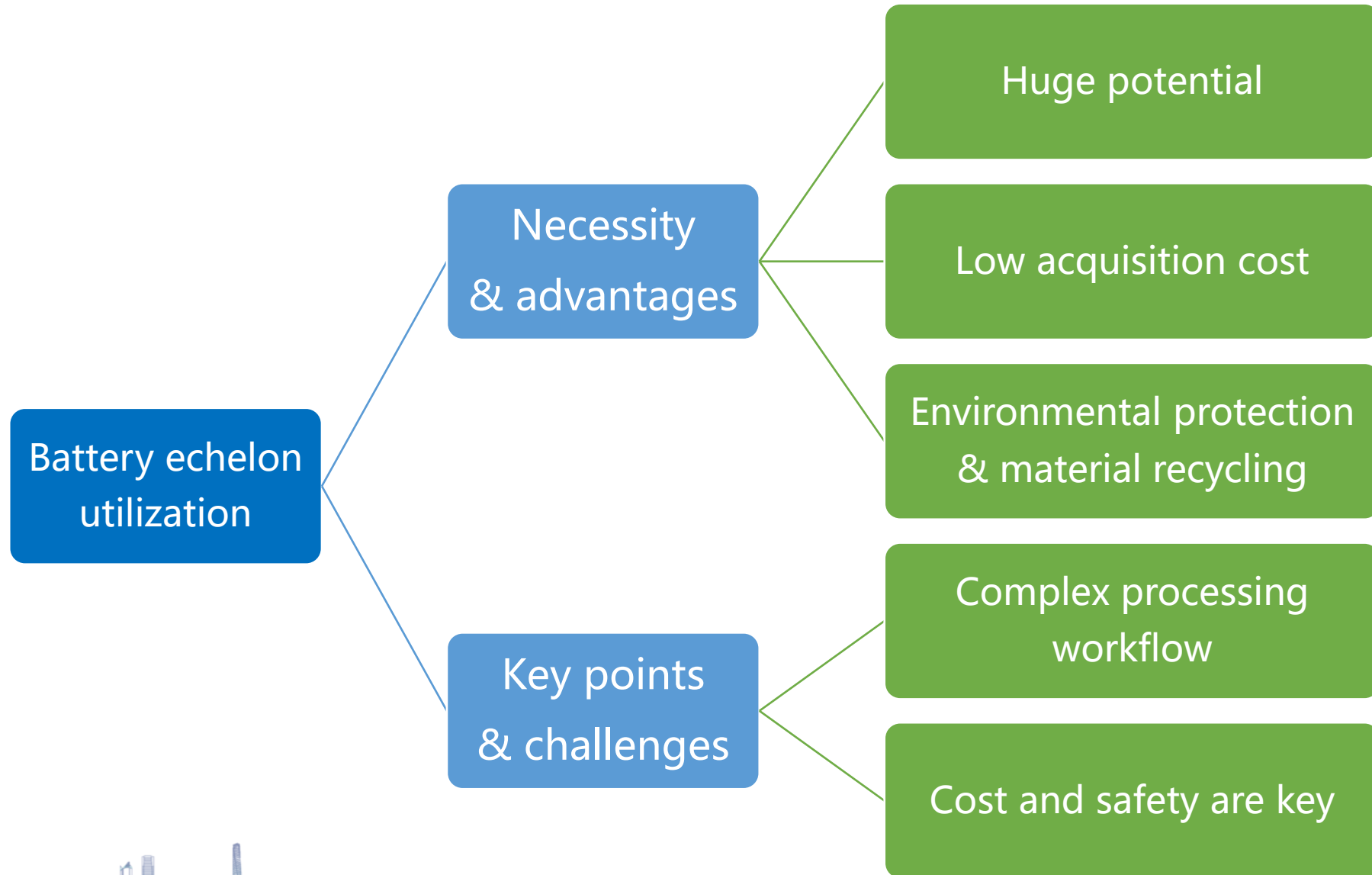
- Keep voltage within acceptable limits
- Participate in reactive-power optimization

Reserve

- Spinning reserve/Interruptible load
- Emergency power supply



Second-Life Utilization of Retired EV Batteries



Part 4

Application of Energy Storage: Challenges and Prospects



Summary and Outlook

Current development and application of new-type storage

- ✓ **New-type storage**, led by lithium batteries, has developed rapidly, with costs falling sharply and technology steadily improving;
- ✓ **Large-capacity storage** applications, led by compressed air and flow batteries, have also achieved breakthroughs;
- ✓ China's **pumped-hydro storage** installed capacity is also growing rapidly, with large approved, under-construction, and planned project capacities;
- ✓ Some large **independent storage** stations and **unit-plus-storage** joint projects have already achieved good returns;
- ✓ As the electricity market develops, storage can now compete in multiple **electricity markets**.



Summary and Outlook

Challenges and outlook

- ✓ The **overall costs** of lithium batteries, flow batteries, and compressed air storage need to be reduced further;
- ✓ Both the policies and the technologies for energy storage stations to participate in electricity markets need further strengthening;
- ✓ The investment and operation model for **grid-side storage** needs further development;
- ✓ A unified design for storage **capacity market/compensation** is anticipated to promote the healthy development of new-type storage;
- ✓ The technology pathway for large-capacity, **long-duration storage** remains unclear and may become key to building the new-type power system in the future.





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